1 FORMAL DEFINITION: INPUT AND OUTPUT

Let $\mathcal{U}$, $\mathcal{B}$ and $\mathcal{M}$ be the sets of users, books and movies, where each user u has a book history $H_b(u) \subseteq \mathcal{B}$ and a movie history $H_m(u) \subseteq \mathcal{M}$. Our solution is a function f that takes a user–book pair (u, b) as input, where u is a cold-start user ($|H_b(u)| < T$ for a threshold T), and outputs a relevance score $f(u, b) \in \mathbb{R}$ for the candidate book b; a higher score means the user is more likely to enjoy the book. Ranking candidate books by this score gives the recommendations.

2 FORMAL DEFINITION: ALGORITHM

Data. The raw data are triples $M' = \{(u, m, r)\}$ and $B' = \{(u, b, r)\}$ with $u \in \mathcal{U}$, $m \in \mathcal{M}$, $b \in \mathcal{B}$ and rating $r \in \{1, \ldots, 5\}$. We build sparse matrices R_M and R_B with users as rows, items as columns and observed ratings as entries (missing entries are not treated as 0). The overlap set is $\mathcal{O} = \{u \in \mathcal{U} \mid H_b(u) \neq \emptyset \wedge H_m(u) \neq \emptyset\}$, with true overlap ratio $\rho_{true} = |\mathcal{O}|/|\mathcal{U}|$. Before any training, we fix a random test set $\mathcal{O}_{test} \subseteq \mathcal{O}$ (20%) and remove its book ratings from B', so these users are cold-start ($T = 1$) and their removed ratings are used only as evaluation ground truth. Let $\mathcal{O}_{train} = \mathcal{O} \setminus \mathcal{O}_{test}$.

Embeddings. Using all users and ratings in each domain, we factorize $R_M \approx U'V'^{\top}$ and $R_B \approx U''V''^{\top}$, where $U', U'' \in \mathbb{R}^{|\mathcal{U}|\times k}$, $V' \in \mathbb{R}^{|\mathcal{M}|\times k}$, $V'' \in \mathbb{R}^{|\mathcal{B}|\times k}$ and k is the embedding dimension. Rows are embeddings: $f_m(u) = U'[u]$, $f_b(u) = U''[u]$ and $v_b = V''[b]$, so $U'[u]\cdot V'[m]$ and $U''[u]\cdot V''[b]$ approximate the rating user u gives movie m or book b. The matrices start as small random values ($\mathcal{N}(0, 0.01)$) and are trained by mini-batch gradient descent on the regularized squared error over observed ratings only:

$$\ell_{P,Q}(u, i, r) = (r - P[u]\cdot Q[i])^2 + \lambda\big(\|P[u]\|^2 + \|Q[i]\|^2\big) \quad (1)$$

$$(U', V') = \arg\min \sum_{(u,m,r)\in M'} \ell_{U',V'}(u, m, r) \quad (2)$$

$$(U'', V'') = \arg\min \sum_{(u,b,r)\in B'} \ell_{U'',V''}(u, b, r) \quad (3)$$

Matrix factorization [2] yields dense vectors whose dot product predicts a rating, which is the representation the mappings operate on. The penalty λ discourages large embeddings that overfit. Hyperparameters: k, λ, learning rate, batch size and epochs.

Overlap subsets. From $\mathcal{O}_{train}$ we draw, by uniform sampling without replacement, $\mathcal{O}_{10}, \mathcal{O}_{20}, \mathcal{O}_{50}, \mathcal{O}_{100}$ containing 10%, 20%, 50% and 100% of its users, simulating different overlap conditions relative to ρ_{true}. Every mapping uses the same subsets, and $\mathcal{O}_{test}$ is never used for training.

Mappings. With embeddings fixed, a mapping $g : \mathbb{R}^k \to \mathbb{R}^k$ translates a user's movie embedding into the book embedding space. For each subset $\mathcal{O}_\rho$, g is trained by gradient descent to minimize

$$L_g = \sum_{u\in\mathcal{O}_\rho} \|g(f_m(u)) - f_b(u)\|^2 \quad (4)$$

and each of the three strategies is trained on each subset (12 mappings). Following the embed-then-map idea of EMCDR [3], all three strategies share this objective, so they differ only in how many users share a set of mapping parameters.

Shared. One multilayer perceptron with a single parameter set, used for every user:

$$g_{shared}(x) = W_2\,\sigma(W_1 x + b_1) + b_2 \quad (5)$$

with $W_1 \in \mathbb{R}^{h\times k}$, $W_2 \in \mathbb{R}^{k\times h}$, hidden width h and ReLU σ. This is the shared mapping of EMCDR [3]; the nonlinearity lets it capture translations a single linear map cannot.

Cluster. K-means partitions $\{f_m(u) : u \in \mathcal{O}_\rho\}$ into clusters $C_1, \ldots, C_K$ with centroids c_j. One network g_j of the same form is trained per cluster, using only the users in C_j. A cold-start user u uses g_{j^*} with $j^* = \arg\min_j \|f_m(u) - c_j\|$. This is inspired by the cluster-level mapping of MacCDR [1], simplified to K-means with independent networks; it assumes users with similar movie taste translate to book taste similarly.

Specific. A meta-network $m_\phi : \mathbb{R}^k \to \mathbb{R}^{k\times k}$ outputs a user-specific matrix $W_u = m_\phi(f_m(u))$, so $g_{specific}(f_m(u)) = W_u f_m(u)$. Only ϕ is trained, using L_g; no per-user mapping is stored. This is inspired by the meta-network of PTUPCDR [4], simplified to take the user's embedding as input.

Mapping hyperparameters: h, σ, learning rate, batch size, epochs, K and the width of m_ϕ.

Scoring. For $g \in \{g_{shared}, g_{cluster}, g_{specific}\}$,

$$f(u, b) = \begin{cases} g(f_m(u)) \cdot v_b & |H_b(u)| < T \\ f_b(u) \cdot v_b & \text{otherwise.} \end{cases} \quad (6)$$

For a cold-start user we score every candidate book, sort by score in descending order and recommend the top-ranked books.

REFERENCES

[1] Yixin Chen, Tianzi Zang, and Juan Li. 2026. MacCDR: A Memory-Augmented Cluster-Level Preference Mapping Framework for Cross-Domain Cold-Start Recommendation. In *Proceedings of BESC 2025, LNCS 16432*. 193–207.

[2] Yehuda Koren, Robert Bell, and Chris Volinsky. 2009. Matrix Factorization Techniques for Recommender Systems. *Computer* 42, 8 (2009), 30–37.

[3] Tong Man, Huawei Shen, Xiaolong Jin, and Xueqi Cheng. 2017. Cross-Domain Recommendation: An Embedding and Mapping Approach. In *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence (IJCAI-17)*. 2464–2470.

[4] Yongchun Zhu, Zhenwei Tang, Yudan Liu, Fuzhen Zhuang, Ruobing Xie, Xu Zhang, Leyu Lin, and Qing He. 2022. Personalized Transfer of User Preferences for Cross-domain Recommendation. In *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining (WSDM '22)*. 1507–1515.